Stage 11E8a Closeout and Regression Closure

Purpose

Stage 11E8a-S0 through S4 complete the mandatory single-source pilot execution. This closeout stage does not add a new scientific inference. It closes the engineering boundary exposed by the package-wide review so that the pilot may be retained as a scientifically partial result without leaving known regression or test-environment defects.

Normative boundaries

Deterministic LD6 research budgets

`MultilevelResearchOptions` is a research profiler contract, not a production scene-admission contract. Its documented defaults are deterministic:

```
max_profile_nodes       4,000,000, additionally clamped by active memory
max_phase_evaluations   2,000
max_workspace_bytes     512,000,000, additionally clamped by active memory
```

The default phase count must not vary with host calibration. An explicit lower or higher positive `max_phase_evaluations` is authoritative for the research run. Production density preparation remains governed by `FrameworkDynamicsResources` and `DensityPlanningLimits`.

Exact explicit-grid Phase-A bounds

When an atomic or framework density channel supplies `grid_shape`, Phase A must use that exact logical node count. It must not substitute the per-field maximum voxel allowance. The explicit-grid path is exact and must not fail on a fictitious maximum-sized mesh. Interval-derived grids retain conservative budget bounds because adaptive refinement can request a finer grid during Phase B. Sparse and automatic Phase-A plans likewise remain budget-bounded when no explicit logical shape is available.

Resource-test isolation

Tests of one hard limit must construct inputs that remain below all earlier hard limits. Runtime-derived guardrails remain authoritative in production; tests may not relax them through legacy count arguments. Test fixtures therefore use a sufficiently large primary memory budget and reduce unrelated Phase-A counts when testing a small scene-wide peak limit.

Optional interactive dependencies

Tests whose scientific operation requires `fast-simplification` must skip at module collection when that optional dependency is absent. Production calls continue to raise `GraphUnsupportedFeatureError` with the installation instruction.

Canonical resource names

Diagnostics use the public `max_density_*` names. Compatibility text may include the backend-local alias, but tests and documentation must assert the public name.

Acceptance criteria

1. The previously failing multilevel, planning, framework-density, and mesh-simplification subset has no failures in a base installation.
2. Explicit `8 x 8 x 8` dense fields record 512 stencil values and do not inherit a fictitious per-field maximum mesh bound.
3. Framework edge quadrature overflow is reported before floating realization.
4. Forced sparse-storage overflow reports `max_density_stored_block_values`.
5. Stage 11E8a-S0 through S4 focused tests remain green.
6. A broader bounded regression sweep records any remaining failures separately from optional-dependency skips.

Scientific status

This closeout does not alter the S4 conclusion. The Na-LTA NVE-continuation pilot remains `scientifically_partial`: occupied basin identities are supported, while PMF force-density agreement and observed transition paths remain blocked by provenance and unresolved transition topology.